

FMDiffBTS: A Lightweight Frequency-decoupled Mamba Diffusion Network for 3D Brain Tumor Segmentation

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Abstract—Diffusion models can deliver high-quality medical image segmentation via iterative denoising, but multi-step inference often incurs prohibitive latency. We propose FMDiffBTS, a lightweight frequency-decoupled Mamba diffusion network for 3D brain tumor segmentation. Specifically, a Frequency-Decoupled Dual-Stream (FD-DS) Encoder leverages 3D discrete wavelet transform (3D-DWT) to decouple low-frequency anatomical structure from high-frequency boundary cues during downsampling. The low-frequency branch adopts Mamba state-space modeling for efficient long-range dependency capture, while the high-frequency branch employs Hybrid Pooling Attention to enhance boundary cues. Moreover, a Multi-scale Axial Fusion Module (MAFM) aggregates multi-resolution decoder features through grouped convolutions and refines tumor structures using separable axial large-kernel convolutions. Experiments on BraTS2020 demonstrate that FMDiffBTS achieves 86.23% mean Dice and 2.45 mm mean HD95 with only 25.9M parameters, outperforming recent baselines under a compact model budget. These results indicate that frequency decoupling with efficient sequence modeling can make diffusion-based 3D segmentation more practical for resource-constrained deployment.

Keywords—Diffusion Model; Brain Tumor Segmentation; Mamba; Lightweight Network

I. INTRODUCTION

Brain tumors are among the most aggressive malignancies of the central nervous system, and accurate segmentation of tumor subregions from multimodal magnetic resonance imaging (MRI) is essential for diagnosis, treatment planning, and longitudinal assessment [1]. However, this task remains challenging due to large inter-patient variability, heterogeneous appearance across modalities, and ambiguous boundaries between lesions and normal tissues in three-dimensional volumes [2]. Manual delineation is time-consuming and prone to inter-observer variability, which motivates reliable automated methods for clinical practice [3].

With the rapid development of deep learning, 3D U-Net has become a strong baseline for volumetric medical image segmentation [4]. To better capture global context, Transformer-based architectures introduce self-attention and achieve competitive performance; however, high-resolution 3D volumes may still impose non-trivial latency and memory cost [5][6]. Recently, state-space models (SSMs), especially Mamba, have attracted increasing attention as efficient alternatives, offering linear-time long-range modeling and promising accuracy–efficiency trade-offs in medical image analysis [7][8].

In parallel, diffusion models have emerged as a powerful paradigm by formulating segmentation as conditional generation through iterative denoising, which is beneficial for recovering fine-grained structures and uncertain boundaries [9][10]. Nevertheless, diffusion-based segmentation typically suffers from excessive computational overhead, since multi-step reverse denoising requires repeated forward passes and thus amplifies inference latency—hindering deployment in resource-constrained clinical environments.

To address these issues, we propose FMDiffBTS, a lightweight frequency-decoupled Mamba diffusion network for 3D brain tumor segmentation. We adopt a Frequency-Decoupled Dual-Stream Encoder that leverages 3D-DWT to disentangle low-frequency structural content from high-frequency boundary, modeled by Mamba layers and Hybrid Pooling Attention, respectively. In addition, a Multi-scale Axial Fusion Module efficiently aggregates multi-resolution decoder features and refines structures with separable axial large-kernel convolutions. Overall, FMDiffBTS balances segmentation accuracy and inference efficiency with a compact parameter budget.

II. METHODS

A. Overall architecture

As shown in Figure 1, FMDiffBTS employs a UNet-like encoder–decoder architecture within a diffusion framework for 3D brain tumor segmentation. The network takes concatenated



Figure 1. Overview of our method.

four-modality MRI images and three noisy label channels as input, with timestep embeddings injected through linear projections. The encoder employs a Frequency-Decoupled Dual-Stream design through three downsampling stages, where a 3D Discrete Wavelet Transform decomposes features into low-frequency structural content (processed by Mamba layers) and high-frequency boundary cues (enhanced by Hybrid Pooling Attention), fused at each scale. The decoder mirrors this structure with three upsampling stages and skip connections. Multi-resolution decoder features (48^3 , 24^3 , 12^3) are aggregated via the Multi-scale Axial Fusion Module using grouped convolutions and refines boundaries via separable axial large-kernel convolutions (kernel sizes 3, 5, 7), achieving efficient segmentation with only three network layers.

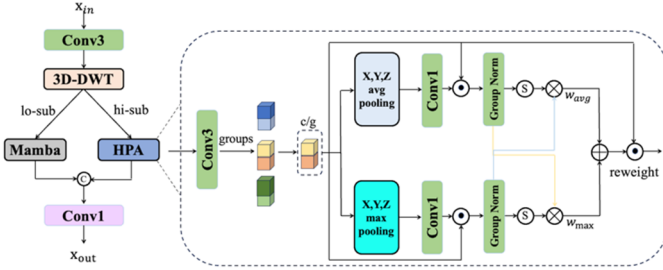


Figure 2. The illustration of FD-DS Encoder.

B. Frequency-Decoupled Dual-Stream Encoder

To mitigate information loss during downsampling, we propose a frequency-decoupled dual-stream encoding strategy, as depicted in Figure 2. At each encoder stage, three-dimensional Discrete Wavelet Transform (3D-DWT, Haar wavelet) decomposes aligned features $\tilde{X}^{(s)} \in \mathbb{R}^{C \times D \times H \times W}$ into frequency subbands:

$$L^{(s)}, \{H_k^{(s)}\}_{k=1}^7 = \text{DWT}_{3D}(\tilde{X}^{(s)}) \quad (1)$$

where $L^{(s)} \in \mathbb{R}^{C \times \frac{D}{2} \times \frac{H}{2} \times \frac{W}{2}}$ captures low-frequency structural semantics (LLL coefficient) and $\{H_k^{(s)}\}$ encode seven high-frequency boundary details (LLH, LHL, LHH, HLL, HLH, HHL, HHH). This decomposition achieves resolution reduction while explicitly preserving frequency-domain information that traditional pooling degrades.

The low-frequency branch leverages stacked Mamba layers to capture long-range dependencies with linear complexity, with mamba_depth set to (1,1,2) across stages:

$$L_{\text{out}}^{(s)} = \text{Mamba}(L^{(s)}, t_{\text{emb}}) \quad (2)$$

where timestep embeddings t_{emb} guide diffusion-aware feature learning, prioritizing global structural consistency across multi-step inference. Unlike Transformer-based methods with quadratic complexity that struggle with high-resolution volumes, Mamba efficiently processes entire 96^3 feature maps through selective state-space modeling, enabling stable capture of anatomical structures during iterative denoising.

For the high-frequency branch, we first aggregate the seven subbands through shared grouped convolutions to reduce computational overhead, yielding $H_{\text{mix}}^{(s)}$. To amplify boundary and texture responses, we introduce Hybrid Pooling Attention (HPA) with the group number set to 4, which captures complementary spatial statistics. Specifically, average pooling (AP) extracts smooth regional cues while max pooling (MP) highlights salient edge activations. The attention mechanism fuses both pooling modes along the depth, height, and width axes:

$$H_{\text{out}}^{(s)} = \sigma \left(\text{Conv} \left(\text{AP}_{\text{DHW}}(H_{\text{mix}}^{(s)}) + \text{MP}_{\text{DHW}}(H_{\text{mix}}^{(s)}) \right) \odot H_{\text{mix}}^{(s)} \right) \quad (3)$$

where $\sigma(\cdot)$ is sigmoid activation, $\odot$ indicates element-wise multiplication, and the gate adaptively highlights informative high-frequency regions. Both branches are merged by channel concatenation followed by a $1 \times 1 \times 1$ convolution:

$$Y^{(s)} = \text{Conv}_{1 \times 1}([L_{\text{out}}^{(s)}, H_{\text{out}}^{(s)}]) \quad (4)$$

This frequency-decoupled design explicitly separates structural and boundary modeling, enabling the diffusion model to preserve critical edge details while maintaining global morphological coherence throughout the reverse sampling process.

C. Multi-scale Axial Fusion Module

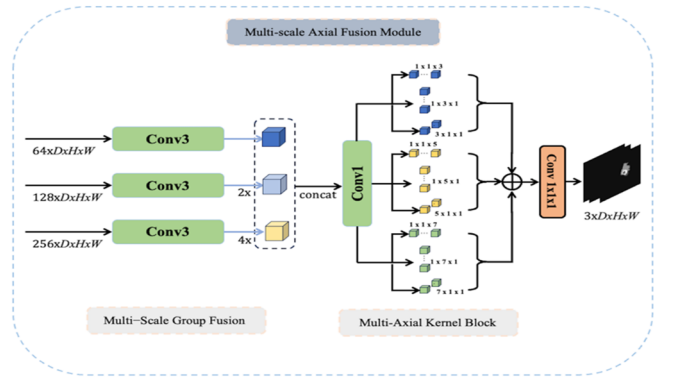


Figure 3. The illustration of MAFM.

The decoder generates multi-scale features at three resolutions (48^3 , 24^3 , 12^3) through progressive upsampling, each capturing structural information at different granularities. To effectively integrate these hierarchical representations for boundary refinement, we propose a Multi-scale Axial Fusion Module, as illustrated in Figure 3, that aggregates features across scales and applies axial convolutions for spatial structure enhancement.

MAFM consists of two components: Multi-Scale Group Fusion (MSGF) and Multi-Axial Kernel Block (MAKB). Given decoder features $\{F_i\}_{i=1}^3$ at different scales, MSGF first aligns channel dimensions through grouped convolutions, then upsamples all features to the highest resolution (96^3):

$$\hat{F}_i = \text{Upsample}(\text{Group Conv}(F_i)), \quad i \in \{1, 2, 3\} \quad (5)$$

where grouped convolutions reduce computational cost while preserving inter-channel dependencies, and the upsampling is performed via trilinear interpolation. The aligned features are concatenated and compressed via $1 \times 1 \times 1$ convolution to generate the fused representation F_{fused} . Subsequently, MAKB applies separable axial large-kernel convolutions with kernel sizes 3, 5, and 7 along depth, height and width axes to capture multi-scale structural patterns:

$$F_{\text{out}} = \text{Conv}_{1 \times 1}(\text{Concat}[\text{AxialConv}_k(F_{\text{fused}})]_{k \in \{3, 5, 7\}}) \quad (6)$$

where axial decomposition reduces the computational complexity of 3D large-kernel convolutions from $O(k^3)$ to $O(3k)$, enabling efficient modeling of elongated structures such as tumor boundaries. The final output is passed to a segmentation head to yield the denoised estimate $\hat{x}_0 \in \mathbb{R}^{3 \times D \times H \times W}$ for the three tumor subregions (WT, TC, ET).

D. Loss Function

During training, the diffusion model learns to predict the clean segmentation mask x_0 from noisy input x_t at timestep $t \in$

$[1, T]$. Given ground truth x_0 and model prediction $\hat{x}_0 = f_\theta(x_t, I, t)$, we employ a composite loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}}(\hat{x}_0, x_0) + \mathcal{L}_{\text{BCE}}(\hat{x}_0, x_0) + \mathcal{L}_{\text{MSE}}(\hat{x}_0, x_0) \quad (7)$$

The Dice loss maximizes region overlap for three tumor subregions (WT, TC, ET), binary cross-entropy loss provides pixel-wise supervision, and mean squared error loss stabilizes training by regularizing prediction smoothness.

III. EXPERIMENTS

A. Dataset

We evaluate FMDiffBTS on the BraTS2020 dataset [11], which contains 369 multi-modal brain MRI cases with four modalities (T1, T1ce, T2, and FLAIR). Expert delineations define three nested regions—Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET)—with data randomly partitioned into 296 training and 73 test volumes.

For evaluation, we report Dice similarity coefficient (Dice) and 95th-percentile Hausdorff distance (HD95). Dice quantifies the volumetric agreement between prediction and ground truth, whereas HD95 measures boundary discrepancy (lower is better). In addition, we include the number of parameters (Params) to reflect model complexity.

B. Implementation details

FMDiffBTS is built with PyTorch 2.0.1 and CUDA 11.7, and all experiments are run on an NVIDIA RTX 4090 GPU (24 GB). We optimize the network for 300 epochs using AdamW with an initial learning rate of $1e-4$, weight decay of $1e-5$, and a cosine-annealing schedule. During training, MRI volumes are randomly cropped into $96 \times 96 \times 96$ patches with a batch size of 2. Data augmentation includes random flipping, rotation, intensity scaling, and elastic deformation. For inference, we adopt DDIM sampling with 10 denoising steps and apply sliding-window prediction with 50% overlap over the full volume.

TABLE I. PERFORMANCE COMPARISON ON BRATS2020 DATASET

Methods	Params(M)	Dice(%) $\uparrow$				HD95(mm) $\downarrow$			
		WT	TC	ET	Mean	WT	TC	ET	Mean
Swin-UNETR[6]	248.8	91.38	84.64	80.34	85.45	2.29	2.83	3.12	2.75
SegFormer3D[5]	99.9	90.73	83.51	78.92	84.39	4.31	3.57	3.63	3.84
VTUNet[12]	47.2	90.32	83.17	77.56	83.68	4.75	4.12	4.76	4.54
SegMamba[8]	269.7	91.92	85.61	79.66	85.73	1.89	2.62	3.35	2.62
EM-Net[13]	219.58	91.97	84.79	79.54	85.43	2.45	3.67	3.68	3.27
Diff-UNet[14]	153.6	91.77	85.16	78.86	85.26	2.15	4.43	3.11	3.23
DiffBTS[15]	27	91.88	85.56	79.55	85.66	2.23	3.35	3.46	3.01
Proposed	25.9	92.23	86.33	80.12	86.23	1.74	2.39	3.22	2.45

C. Comparative Experiment

We benchmark FMDiffBTS against seven representative brain tumor segmentation approaches, covering Transformer-based models (Swin-UNETR [6], SegFormer3D [5], VTUNet [12]), Mamba-based methods (SegMamba [8], EM-Net [13]), and diffusion-based baselines (Diff-UNet [14], DiffBTS [15]).

According to Table I, FMDiffBTS attains the top overall accuracy, achieving 86.23% mean Dice and 2.45 mm mean HD95. Notably, the improvements are obtained under the same evaluation protocol for all methods, and our diffusion inference uses 10 DDIM sampling steps to balance accuracy and efficiency. We attribute the consistent gains to the proposed frequency-decoupled encoder that preserves boundary cues

during downsampling, together with the multi-scale axial fusion head that better aggregates hierarchical decoder features for refinement. Qualitative visualizations are shown in Figure 4, where two cases are randomly selected for comparison across different methods. Relative to DiffBTS, which has a comparable parameter budget, our method increases mean Dice by 0.57% and lowers mean HD95 by 0.56 mm. For the challenging TC and ET regions, FMDiffBTS reaches Dice scores of 86.33% and 80.12%, surpassing DiffBTS by 0.77% and 0.57%, respectively. In addition, when compared with lightweight baselines such as SegFormer3D, FMDiffBTS still shows stronger overall performance on both mean Dice and mean HD95.

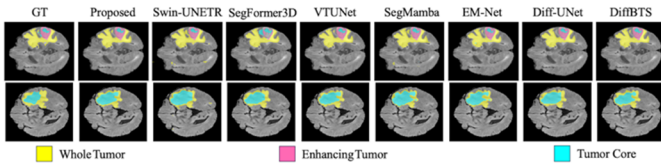


Figure 4. Qualitative comparisons of segmentation results.

TABLE II. THE RESULTS OF ABLATION STUDY

Methods	Dice(%) $\uparrow$			Params(M)
	WT	TC	ET	
Basic	91.27	85.01	78.26	23.96
Basic+FD-DS Encoder	91.63	86.06	79.51	24.12
Basic+MAFM	91.77	85.98	78.58	25.74
Basic+FD-DS Encoder+MAFM	92.23	86.33	80.12	25.9

D. Ablation Experiment

To assess the contribution of each module, we perform ablation studies on BraTS2020. As shown in Table II, we evaluate four configurations: Basic, Basic + FD-DS Encoder, Basic + MAFM, and the full model. The FD-DS Encoder improves Dice scores to 91.63% (WT), 86.06% (TC), and 79.51% (ET) with only 0.16M additional parameters, verifying that DWT-based frequency decomposition and Mamba processing enhance feature representation under a lightweight budget.

The full model achieves the best results with Dice scores of 92.23% (WT), 86.33% (TC), and 80.12% (ET), bringing gains of +0.96%, +1.32%, and +1.86% over the baseline. These results demonstrate that frequency-decoupled encoding and multi-scale axial fusion complement each other for accurate tumor delineation.

IV. CONCLUSION

In this paper, we proposed FMDiffBTS, a lightweight frequency-decoupled Mamba diffusion framework for 3D brain tumor segmentation. We introduce a Frequency-Decoupled Dual-Stream encoder with 3D-DWT and an efficient Multi-scale Axial Fusion Module, our method effectively captures low-frequency structural semantics and high-frequency boundary cues while maintaining a compact model budget. Experiments on BraTS2020 demonstrate that FMDiffBTS achieves 86.23% mean Dice and 2.45 mm mean HD95 with only 25.9M parameters, showing a favorable accuracy–efficiency trade-off compared with recent strong baselines. In the future, we will further explore faster sampling strategies and broader validation

on additional clinical datasets to improve generalization and practical deployment.

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